

“No new development is happening using parallel bus architecture”



Paul Dhillon, president and chief executive officer of VTI Instruments (formerly known as VXI Technology), talks to EFY's Shweta Dhadiwal Baid about modular test and measurement instrumentation. During the conversation, he shared technology trends that are shaping up the test and measurement industry, and how India is becoming a strategic market for defence and aerospace industries

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How has the test and measurement instrumentation evolved over the years?

Advancement in technology is very well reflected and supported by changes in test and measurement instrumentation. The move to high-speed serial from parallel bus architectures (PCI, VME, PXI, VXI and GPIB) has become prevalent, to the point that no new development is happening using parallel bus architectures. Even embedded chip makers like Intel and Freescale (formerly Motorola) have adopted switched serial buses. Ethernet is an example of a switched serial bus, which is forward-looking and backward-compatible.

In 2005, VXI Technology (now VTI Instruments) and Agilent co-founded LXI (LAN eXtensions for Instrumentation) standard based on Ethernet. It allows T&M manufacturers to replace existing parallel bus architectures with

Ethernet. All the large manufacturers worldwide (including Tektronix, Rohde & Schwartz, Agilent Technologies and VTI Instruments) are part of the LXI consortium and developing products on LXI.

What is VME/VXI- and PXI-based instrumentation? How is it different from LXI?

PXI (PCI eXtensions for Instrumentation) and VXI (VME eXtensions for Instrumentation) have historically been the largest open-standard platforms for modular instrumentation. Both of these platforms require card cages, and are used to configure and build automated test systems. PXI is based on the PCI bus architecture and was designed by the *PCI Industrial Computer Manufacturers Group*. VXI is based on the VMEbus architecture and was designed by the *VITA group*.